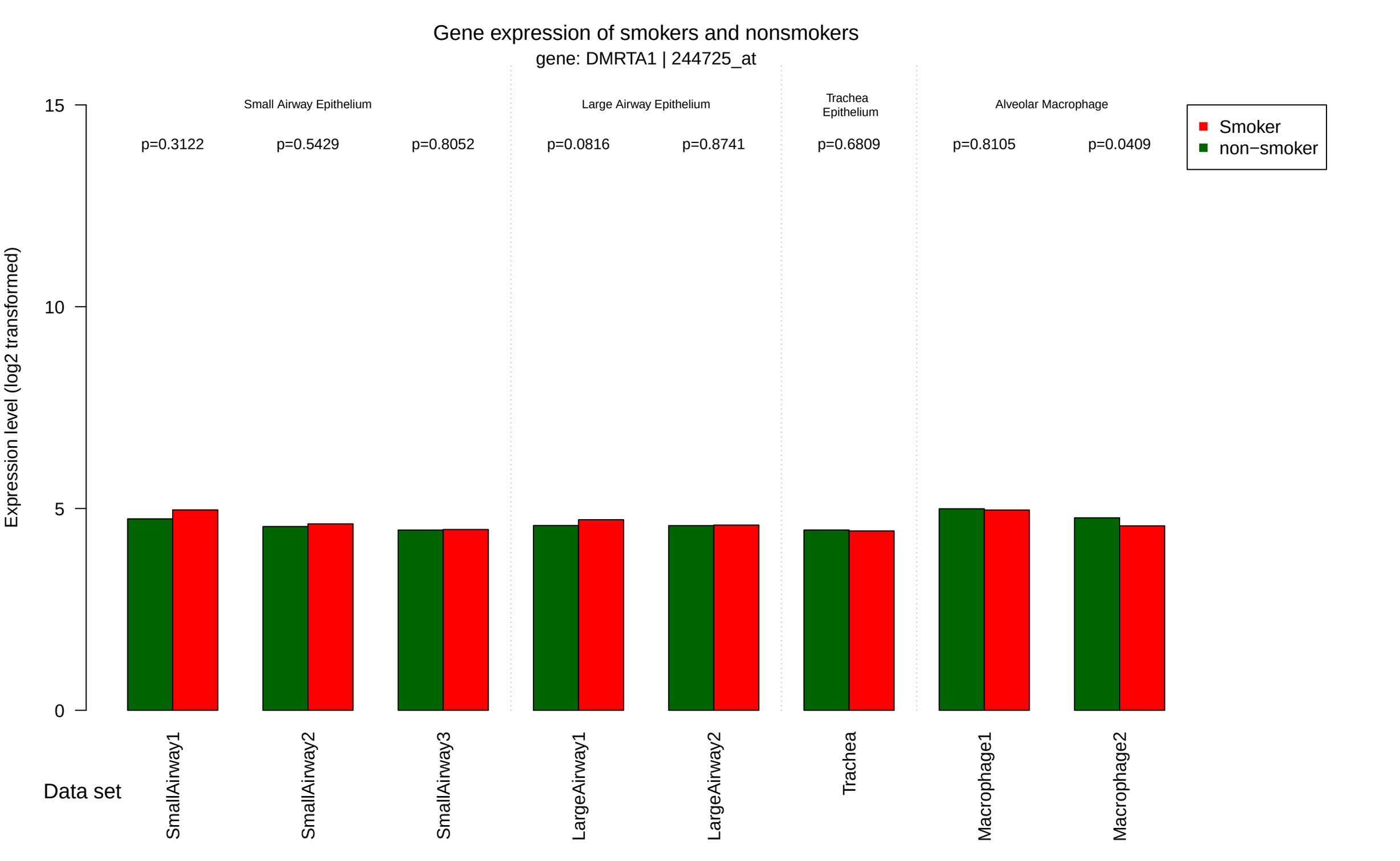
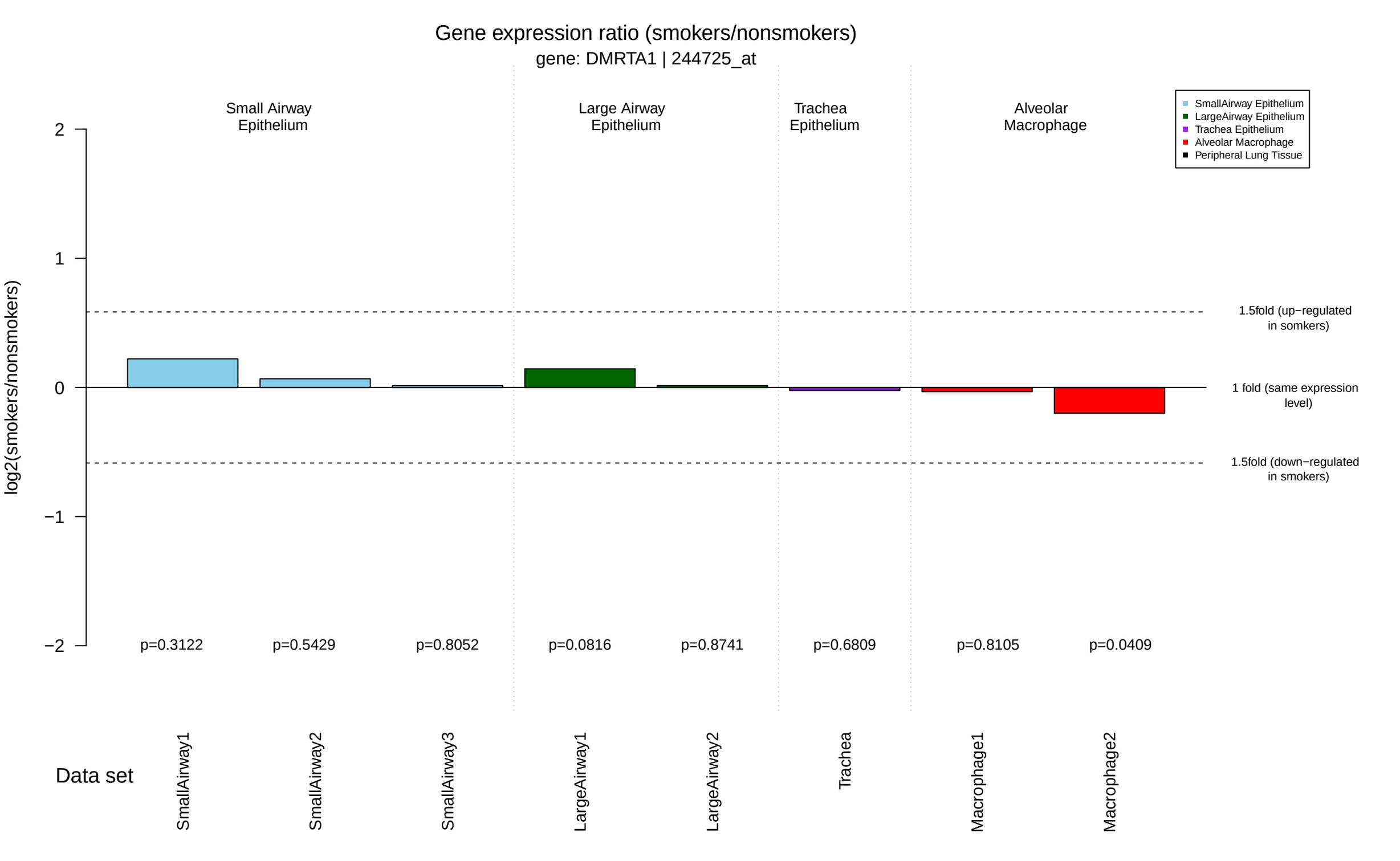
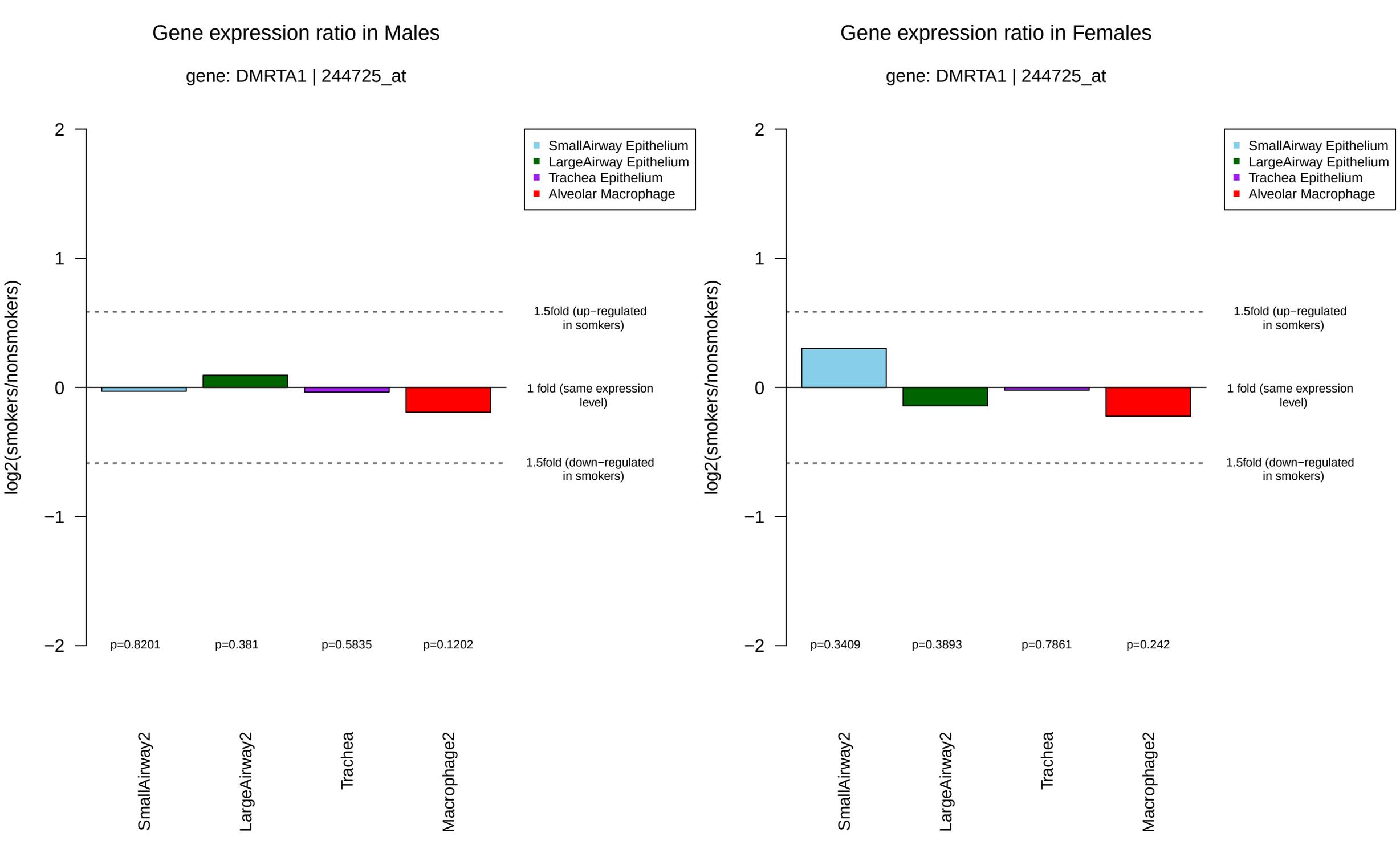
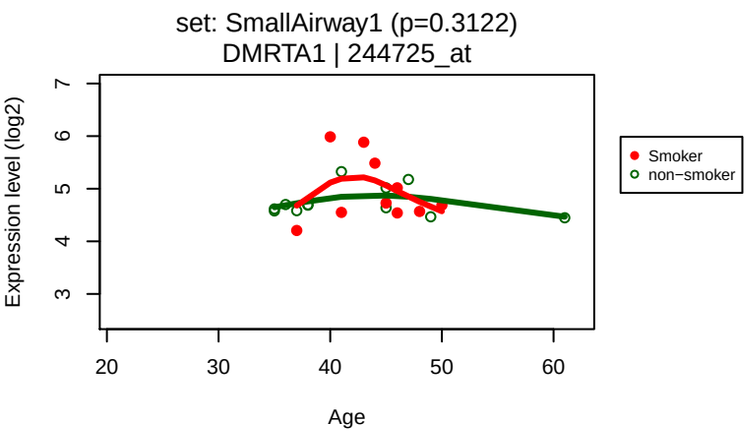
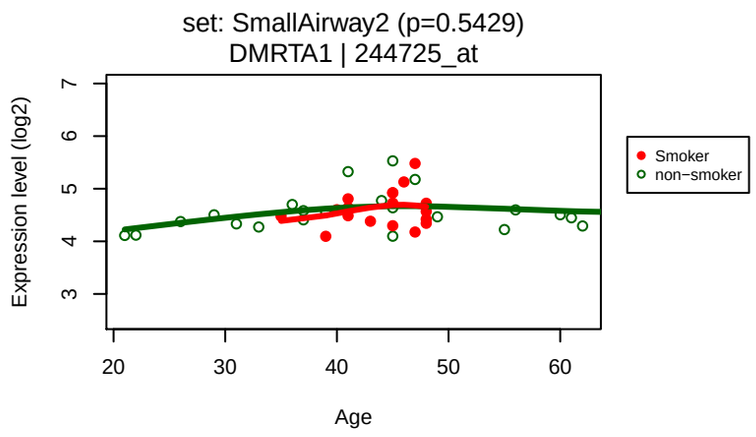
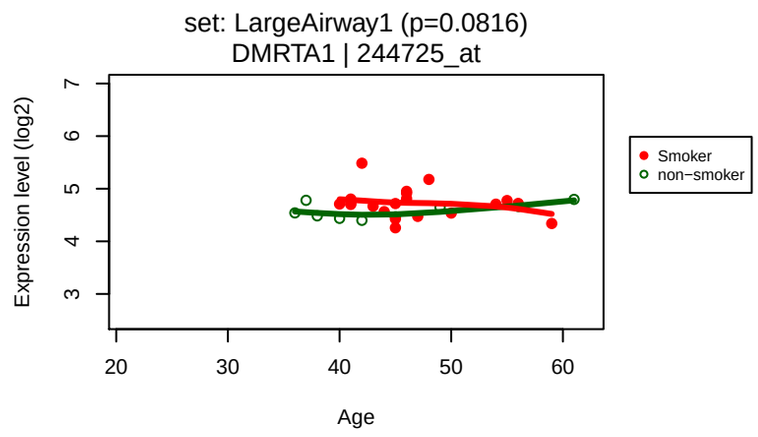

Gene expression of smokers and nonsmokers
gene: DMRTA1 | 244725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3122
p=0.5429
p=0.8052
p=0.0816
p=0.8741
p=0.6809
p=0.8105
p=0.0409
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

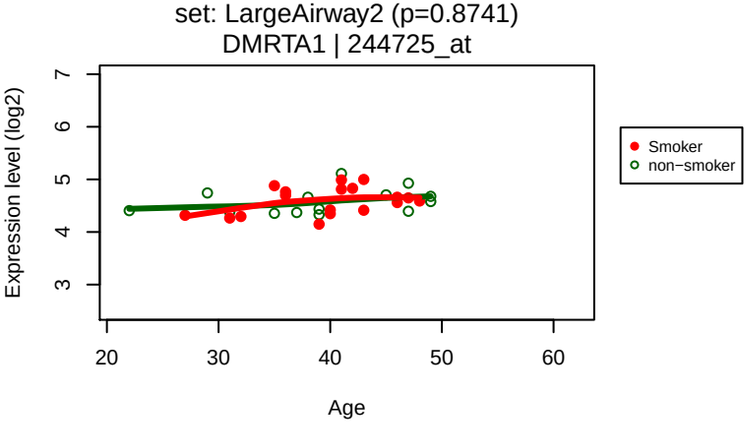

Gene expression ratio (smokers/nonsmokers)
gene: DMRTA1 | 244725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3122
p=0.5429
p=0.8052
p=0.0816
p=0.8741
p=0.6809
p=0.8105
p=0.0409
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

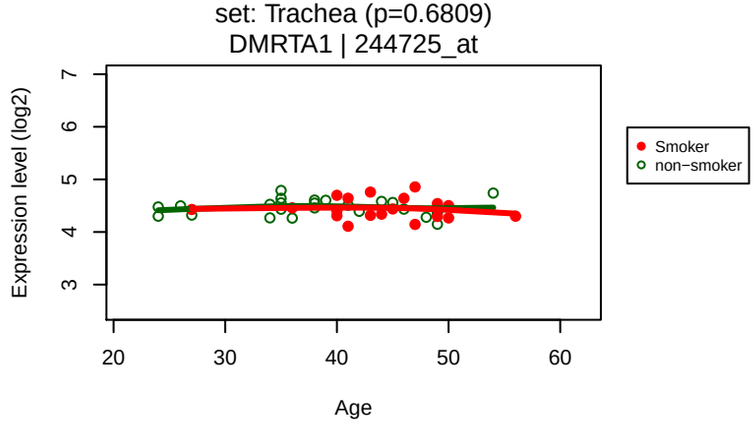

Gene expression ratio in Males
gene: DMRTA1 | 244725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8201
p=0.381
p=0.5835
p=0.1202
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DMRTA1 | 244725_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3409
p=0.3893
p=0.7861
p=0.242
SmallAirway2
LargeAirway2
Trachea
Macrophage2

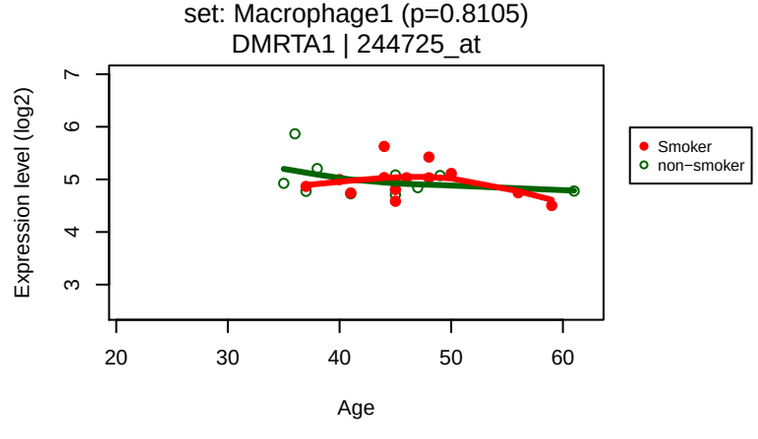

set: SmallAirway1 (p=0.3122)
DMRTA1 | 244725_at
Expression level (log2)
Age
Smoker
non−smoker

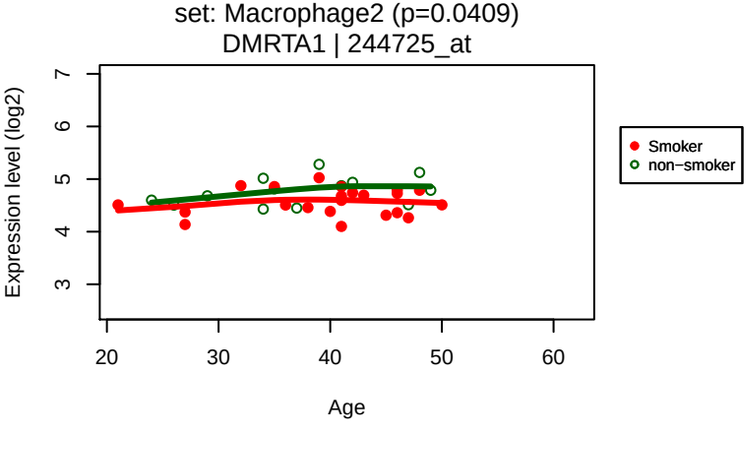

set: SmallAirway2 (p=0.5429)
DMRTA1 | 244725_at
Expression level (log2)
Age
Smoker
non−smoker

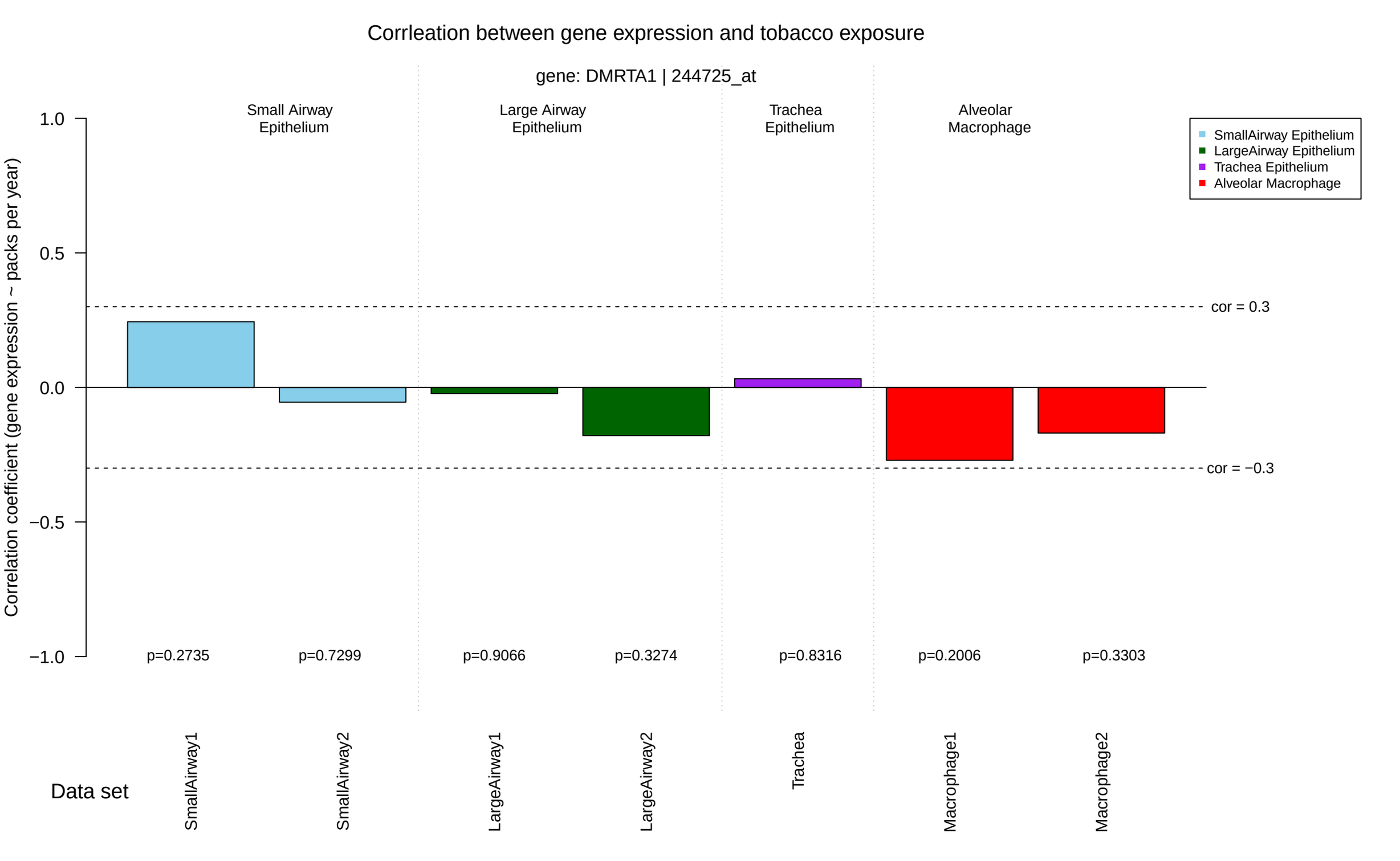

set: LargeAirway1 (p=0.0816)
DMRTA1 | 244725_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8741)
DMRTA1 | 244725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6809)
DMRTA1 | 244725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8105)
DMRTA1 | 244725_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0409)
DMRTA1 | 244725_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DMRTA1 | 244725_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2735
p=0.7299
p=0.9066
p=0.3274
p=0.8316
p=0.2006
p=0.3303
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2